

CarbonLens AI

Mini-project-I Report

Submitted in partial fulfillment of the requirements

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Department of Computer & Science Engineering (AIML)

by

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(Under the ambit of D. Y. Patil Deemed to be University)

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Certificate

This is to certify that, the Mini-project-I titled

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is a bonafide work done by

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degree of

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We declare that this written submission represents our ideas in our own word and where others ideas or words have been included, We have adequately cited and referenced the original source. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of above will be cause for disciplinary action by institute and can also evoke penal action from the source which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Introduction of CarbonLens AI: A sustainability analytics framework The aim of this project is to provide a tool that would help in measuring and optimizing the carbon footprint associated with the training of machine learning models. In today's world, due to the rapid adoption of deep learning and artificial intelligence techniques, machine learning training is becoming increasingly computational and is thus consuming greater amounts of electricity and carbon. While machine learning processes currently emphasize various performance parameters such as accuracy, loss, and speed, little effort has been made to track environmental effects. Framework Description: The framework can be easily integrated into the machine learning training pipeline and will continuously monitor CPU/GPU utilization, run time, and system resources. Using this information, it will calculate the overall energy consumption and convert it to carbon emissions by considering electricity carbon intensity in the region. Model accuracy over each epoch and the Accuracy vs Carbon trade-off curves can be plotted. Furthermore, CarbonLens AI offers sustainability optimization suggestions for your machine learning projects, including early stopping, carbon budget-based training, hyperparameter tuning, and cost reduction on clouds. By using the CarbonLens AI platform, Green AI can be practiced as developers can find a good compromise between performance and sustainability. Keywords: Green AI, Carbon Emission, Sustainability Analytics, Machine Learning, Energy Monitoring

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Chapter 1

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have become essential technologies within modern computing environments to perform intelligent automation, predictions, and decision-making within various industries. Nevertheless, due to the emergence of deep learning architecture and the scale of training, the computing power required to train modern ML models has been growing. As a result, contemporary machine learning algorithms have increased the need for electricity and indirectly contributed to the emission of CO₂ into the atmosphere. Traditionally, during development processes, ML engineers primarily concentrate on optimization of performance metrics such as accuracy, precision, recall, F1, and loss reduction. Nevertheless, while the metrics mentioned above are crucial for ML model's effectiveness, the amount of electricity consumed during its development remains an overlooked issue, with training epochs and retraining contributing to excess energy usage. The necessity to develop sustainable computing solutions has raised the importance of intelligent systems capable of estimating and reducing the carbon footprint of computing activities. This mini project presents CarbonLens AI, a framework developed specifically for analyzing sustainability in ML training environments. Within its functionality, the tool estimates CPU and GPU usage and runtime, calculates the total energy consumption, determines the carbon emission based on local carbon intensity, and produces sustainability insight for developers.

CarbonLens AI ensures that sustainable factors are included in the models' creation by implementing Green AI, which ensures that developers can optimize their models while remaining sustainable.

1.1 Literature Review

Recent research in sustainable AI and green computing highlights the increasing carbon footprint of machine learning model training. Several carbon estimation tools have been developed to address this issue, including CodeCarbon, CarbonTracker, eco2AI, and Experiment Impact Tracker. These systems estimate carbon emissions by monitoring hardware usage and converting the corresponding energy consumption into CO₂ values. Most existing tools focus only on reporting carbon emissions after training completion. They do not provide deeper insights into the relationship between carbon usage and model accuracy progression across epochs. As a result, developers are unable to identify inefficient training stages where energy consumption continues to increase while model improvement remains negligible.

The concept of Green AI, introduced by Schwartz et al., emphasizes the importance of considering efficiency, computational cost, and carbon footprint alongside model performance. Inspired by these limitations and research directions, CarbonLens AI extends existing solutions by integrating sustainability analytics, tradeoff visualization, optimization suggestions, and carbon budget-aware decision support.

1.1.1 Information of Survey

- The survey of existing literature and tools reveals that current carbon estimation frameworks provide basic emission tracking but lack decision-support mechanisms for sustainable training optimization.
- **Methodology of Reference Tools**

The following methodologies are commonly used:

- hardware utilization monitoring
- runtime logging
- average power estimation
- carbon intensity conversion
- experiment-level emission reporting

- **Limitations Identified**

- no Accuracy vs Carbon tradeoff graph
- no sustainability score
- no carbon budget thresholding
- no optimization recommendations
- no explainable ESG reporting

- **Need for CarbonLens AI**

To overcome these limitations, CarbonLens AI introduces:

1. real-time resource monitoring
2. energy estimation during training
3. carbon-aware analytics
4. sustainability optimization suggestions
5. ESG-style reporting framework

This survey strongly supports the need for a unified sustainability analytics framework specifically for machine learning workflows.

1.2 Motivation

- The primary motivation behind this project is the growing environmental impact caused by large-scale machine learning and deep learning training systems.
- Modern AI models are becoming increasingly resource-intensive, often requiring:
 - high-end GPUs
 - long training durations
 - multiple retraining cycles
 - large hyperparameter search spaces
- These processes lead to increased electricity consumption and significant carbon emissions, especially in cloud-based environments.

- Even after model accuracy saturates, training is often continued for additional epochs, causing unnecessary carbon usage and increased cloud costs. Therefore, a system that helps identify low sustainability gain epochs and recommends greener optimization strategies becomes highly valuable.
- CarbonLens AI is motivated by the need to make machine learning development more responsible, efficient, and environmentally sustainable.

1.3 Objective

The primary objective of CarbonLens AI is to develop a sustainability analytics framework that enables carbon-aware machine learning model training.

The specific objectives of the proposed work are as follows:

- To monitor CPU, GPU, memory, and runtime utilization during machine learning model training.
- To estimate total energy consumption based on observed hardware usage and training duration.
- To convert energy consumption into carbon emissions (CO₂) using region-based electricity carbon intensity values.
- To analyze the relationship between model accuracy and carbon emissions throughout the training process.
- To generate an Accuracy vs Carbon Tradeoff Curve for identifying inefficient training epochs.
- To provide optimization recommendations such as early stopping, batch-size tuning, and carbon budget-aware training.
- To support Green AI practices by reducing environmental impact and cloud operational costs.
- To create an ESG-style sustainability report for better decision-making and reporting.

These objectives ensure that machine learning development can be optimized not only for performance but also for sustainability and responsible computing.

1.4 Scope for work

The scope of the proposed mini-project includes the design and implementation of a carbon-aware sustainability analytics framework for machine learning workflows.

The project scope covers the following areas:

- monitoring system resource usage during model training
- energy estimation based on runtime hardware consumption
- carbon emission calculation using regional grid intensity
- tradeoff analysis between model accuracy and carbon footprint
- sustainability-based optimization recommendations
- dashboard-based output visualization
- ESG report generation for sustainability insights
- academic and industrial machine learning experimentation

The framework can be applied in:

- AI research laboratories
- cloud-based machine learning platforms
- sustainable computing initiatives
- enterprise-level model deployment pipelines
- educational deep learning experiments

This scope makes CarbonLens AI highly relevant for both academic research and practical AI system deployment.

1.5 Organization of Report

This report is divided into several chapters to provide a clear and systematic understanding of CarbonLens AI: A Carbon-Aware Sustainability Analytics Framework for Machine Learning Training. Each chapter focuses on a specific aspect of the project, beginning from the problem background and extending to the final framework implementation, results, and future enhancements. The report is organized as follows:

- **Chapter 1: Introduction**

This chapter presents the overview of the CarbonLens AI framework, the motivation behind developing a sustainability-focused machine learning analytics system, and the problem statement. It also highlights the objectives, scope, and significance of integrating carbon-awareness into AI model training workflows.

- **Chapter 2: Literature Survey**

This chapter reviews existing tools and research works related to carbon emission tracking in machine learning systems, such as CodeCarbon, CarbonTracker, and Green AI studies. It discusses the limitations of current frameworks and establishes the need for a more intelligent sustainability analytics system.

- **Chapter 3: CarbonLens AI Framework Design and Methodology**

This chapter explains the proposed framework architecture and the methodology used in the system. It includes the monitoring module, energy estimation engine, carbon conversion engine, sustainability analytics engine, dashboard interface, and ESG report generation workflow.

- **Chapter 4: Implementation, Results and Discussion**

This chapter focuses on the technical implementation details of the CarbonLens AI system. It discusses the technology stack, system modules, output dashboards, evaluation metrics, and the analysis of the Accuracy vs Carbon Tradeoff Curve along with expected observations and benefits.

- **Chapter 5: Conclusion and Future Scope**

The final chapter summarizes the outcomes of the proposed framework, highlights its role in promoting Green AI and sustainable machine learning, and discusses future

improvements such as cloud carbon APIs, multi-region deployment, and advanced optimization using reinforcement learning.

This organized structure helps provide a clear understanding of the complete project development lifecycle, from the initial sustainability problem identification to framework design, implementation, analysis, and future enhancements.

Chapter 2

Literature Survey

The growing computational demand of modern Artificial Intelligence and Machine Learning models has raised serious concerns regarding their environmental impact. Large-scale deep learning architectures, transformer-based models, and repeated experimentation pipelines consume substantial computational resources, leading to high electricity usage and indirect carbon emissions.

In recent years, researchers and developers have started focusing on Green AI, which emphasizes not only model performance but also energy efficiency, carbon footprint reduction, and sustainable deployment practices.

Several existing tools and frameworks have been proposed to estimate carbon emissions during machine learning model training. These systems mainly focus on hardware resource monitoring, power estimation, and carbon intensity-based conversion.

One of the widely used frameworks is **CodeCarbon**, which provides carbon emission estimation for Python-based machine learning workflows. It monitors CPU usage, GPU usage, RAM allocation, and execution time, and then converts the estimated power consumption into CO₂ emissions using country-specific grid intensity values. While it provides useful emission reports, it lacks advanced sustainability analytics such as optimization suggestions and tradeoff visualization.

Another important framework is **CarbonTracker**, which monitors the energy consumption of deep learning experiments and predicts the carbon impact of future training runs. It helps in understanding the ecological cost of model training; however, its focus remains largely limited to tracking and reporting rather than intelligent sustainability decision support.

The **Experiment Impact Tracker** is another tool designed to record the environmental cost of machine learning experiments. It collects runtime information and provides estimates for electricity consumption and CO₂ emissions. However, it does not support real-time tradeoff analysis between performance metrics and sustainability indicators.

Recent research on **eco2AI** has further highlighted the importance of tracking carbon emissions as a first step toward sustainable AI development. The framework emphasizes the need for carbon-aware experimentation but still lacks an integrated system for identifying inefficient epochs and suggesting carbon-efficient optimizations.

The concept of **Green AI**, introduced by Schwartz et al., strongly emphasizes reporting not only model accuracy but also training cost, computational power, and environmental footprint. This concept forms the core inspiration behind CarbonLens AI, as it promotes responsible AI development where efficiency and sustainability are treated as first-class performance indicators.

Although the above tools and research contributions provide strong foundations for carbon estimation, they present several limitations:

- lack of Accuracy vs Carbon tradeoff visualization
- absence of sustainability optimization recommendations
- no carbon budget-aware training support
- no explainable ESG-style sustainability reporting
- limited focus on decision support for overtraining detection
- no intelligent suggestions for low sustainability gain epochs

These limitations create the need for a more comprehensive framework that goes beyond emission tracking and actively helps developers make environmentally responsible training decisions.

To address these gaps, the proposed CarbonLens AI framework integrates:

- real-time resource monitoring
- energy estimation
- carbon conversion

- tradeoff analytics
- optimization recommendations
- sustainability dashboard
- ESG report generation

This makes the proposed system more practical, explainable, and suitable for both academic and industrial machine learning workflows.

2.1 Analysis of Comparative Study

From the above comparative analysis, it is evident that most existing tools primarily focus on carbon emission estimation and experiment-level reporting. While these systems provide valuable insights into energy usage and CO₂ emissions, they do not assist developers in making sustainability-aware optimization decisions during model training.

The major limitations observed in existing frameworks include:

- absence of Accuracy vs. Carbon tradeoff analysis
- no intelligent detection of low sustainability gain epochs
- lack of carbon budget threshold-based training
- no explainable ESG-style sustainability reporting
- limited support for cost-aware cloud optimization

The proposed **CarbonLens AI** framework addresses all these research gaps by integrating sustainability analytics directly into the machine learning training workflow.

By combining resource monitoring, energy estimation, carbon conversion, performance tradeoff visualization, and optimization recommendations, CarbonLens AI extends beyond traditional carbon tracking systems and provides a more intelligent decision-support framework for Green AI.

Tool / Framework	Carbon Tracking	Accuracy vs Carbon Tradeoff	Optimization Suggestions	ESG / Sustainability Reporting
CodeCarbon	Yes	No	No	No
CarbonTracker	Yes	No	Limited	No
Experiment Impact Tracker	Yes	No	No	No
eco2AI	Yes	No	No	Limited
CarbonLens AI (Proposed)	Yes	Yes	Yes	Yes

Figure 2.1: Comparative Analysis of Existing Carbon Tracking Tools

Chapter 3

Methodology

The methodology of CarbonLens AI focuses on integrating sustainability analytics directly into the machine learning training workflow. The proposed framework follows a structured pipeline beginning from model execution and ending with sustainability reporting and optimization recommendations. The methodology is designed to continuously monitor computational resources, estimate energy usage, convert it into carbon emissions, analyze the relationship between model accuracy and environmental impact, and finally generate optimization suggestions for Green AI practices.

3.1 Block Diagram

The block diagram of CarbonLens AI represents the major functional modules involved in carbon aware machine learning training. The complete working of the system can be understood in the following numbered steps:

- 1. User ML Training Script**

The process starts when the user executes a machine learning or deep learning model training script. This acts as the input source for the CarbonLens AI framework.

- 2. CarbonLens Monitoring Layer**

The monitoring layer continuously tracks important system resource metrics such as CPU utilization, GPU utilization, memory usage, runtime duration, and epoch execution time during model training.

- 3. Metrics Logging Module**

The monitored hardware and performance metrics are recorded in structured log files or DataFrames. This ensures that epoch-wise system activity is available for further analysis.

4. Energy Estimation Module

The logged system metrics are processed to estimate the total power consumption and energy usage during the training process.

5. Carbon Conversion Engine

The estimated energy consumption is converted into carbon emissions using region-based carbon intensity factors. This gives the total CO₂ footprint of the training session.

6. Sustainability Analytics Engine

The framework combines carbon emission values with model performance metrics such as accuracy and loss. This module generates sustainability insights like the Accuracy vs. Carbon Tradeoff Curve, carbon efficiency score, and low sustainability gain detection.

7. Optimization Recommendation Module

Based on the sustainability analysis, the framework suggests optimization techniques such as early stopping, epoch reduction, batch-size tuning, and carbon budget-aware training.

8. Dashboard and ESG Report Generator

Finally, all processed outputs are displayed in the dashboard as graphs, metric cards, and reports. ESG-style sustainability reports are also generated for documentation and decision support.

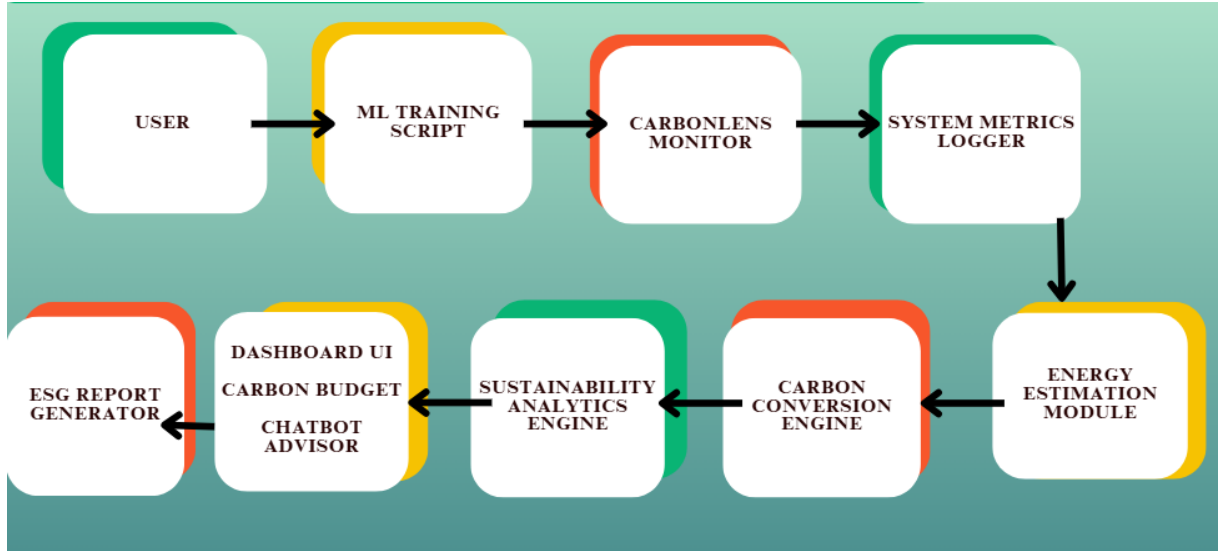


Figure 3.1: Proposed System (Block Diagram)

3.2 System Architecture

The CarbonLens AI framework follows a layered architecture design, where each layer is responsible for a specific analytical responsibility.

The layered architecture improves:

- modularity
- scalability
- maintainability
- deployment flexibility
- easier debugging

The major layers of the proposed system are:

1. Monitoring Layer
2. Analytics Layer
3. Visualization Layer
4. Reporting Layer
5. Optimization Layer

This design ensures smooth data flow from raw resource metrics to final sustainability decisions.

3.3 Layered System Architecture

1. Frontend (Client Side)

The frontend layer acts as the user interaction dashboard for CarbonLens AI. It provides:

- training metrics visualization
- energy usage graphs
- carbon emission cards
- Accuracy vs. Carbon curve
- sustainability score
- optimization alerts
- ESG report preview

This layer can be built using Streamlit and Plotly dashboards.

2. Backend (Server Side)

The backend layer handles:

- metric aggregation
- CSV logging
- DataFrame processing
- sustainability calculations
- report generation
- API communication

Python acts as the primary backend environment.

3. Machine Learning Layer

This layer is responsible for:

- executing ML training scripts
- collecting epoch-level accuracy
- loss tracking
- hyperparameter experimentation
- training lifecycle control

This is the main integration point of CarbonLens AI.

4. Database Layer

The database layer stores:

- epoch logs
- energy metrics
- carbon emissions
- sustainability scores
- optimization history
- experiment metadata

CSV files and Pandas DataFrames act as lightweight storage in the mini-project scope.

5. Security Layer

The security layer ensures:

- protected experiment logs
- safe dashboard access
- report integrity
- controlled data sharing
- secure API keys for cloud carbon APIs (future scope)

3.4 Technology Used

The CarbonLens AI framework uses a combination of monitoring, analytics, visualization, and machine learning technologies.

3.4.1 Frontend Technologies

- Streamlit
- Plotly
- HTML/CSS dashboard components

3.4.2 Backend Technologies

- Python
- Pandas
- psutil
- NumPy

3.4.3 Machine Learning Component

- Scikit-learn
- TensorFlow / PyTorch
- model training scripts
- epoch-level accuracy logging

3.4.4 Database Layer

- CSV files
- Pandas DataFrames
- local experiment storage

3.4.5 Security and Verification Technologies

- secure logging
- access-restricted dashboard

- report checksum validation
- API token protection

3.4.6 Development and Testing Tools

- Jupyter Notebook
- VS Code
- Google Colab
- GitHub
- Streamlit local server

Chapter 4

Result and Analysis

4.1 Analysis

The CarbonLens AI framework was evaluated extensively to analyze its capability in integrating sustainability metrics into machine learning workflows. The evaluation focused on four major aspects: system resource monitoring, energy estimation, carbon emission calculation, and sustainability-aware optimization.

The framework demonstrated effective real-time monitoring of critical hardware resources, including CPU utilization, GPU utilization, memory consumption, runtime duration, and epoch-wise execution time. These metrics were continuously captured during the execution of machine learning training scripts, ensuring fine-grained visibility into the computational behavior of the model. The ability to monitor epoch-level activity enabled the system to perform detailed temporal analysis of resource consumption patterns.

The collected system metrics were then processed by the energy estimation module to calculate the total power consumption during the training process. This estimation was performed using standardized power usage models, ensuring consistency and reliability of results. The computed energy consumption values were further translated into equivalent CO₂ emissions using region-based carbon intensity factors. This conversion provided a meaningful measure of the environmental impact associated with the training process, bridging the gap between computational performance and ecological cost.

One of the most significant analytical contributions of the framework is the generation of the *Accuracy vs. Carbon Tradeoff Curve*. This curve provides a visual and quantitative representation of the relationship between model accuracy and carbon emis-

sions. The experimental results indicate that during the initial epochs, there is a steep improvement in model accuracy with relatively low carbon expenditure. However, as training progresses, the rate of accuracy improvement gradually diminishes, while the carbon emissions continue to increase steadily.

This behavior highlights the concept of diminishing returns in machine learning training. Beyond a certain epoch threshold, additional training results in marginal gains in accuracy but incurs disproportionately higher energy consumption and carbon emissions. The framework successfully identifies these regions as *low sustainability gain epochs*, which are critical for making informed optimization decisions.

The identification of such epochs enables the system to recommend efficient training strategies, such as early stopping and optimal epoch selection. These strategies not only reduce unnecessary computation but also minimize cloud infrastructure costs, making the system economically as well as environmentally efficient. This is particularly important in large-scale deep learning models, where training costs and energy consumption can be significantly high.

In addition to tradeoff analysis, the framework computes several key evaluation metrics that provide a comprehensive view of sustainability performance:

- Total Energy Consumption (kWh)
- Total CO₂ Emissions (kg)
- Model Accuracy (%)
- Carbon Efficiency Score

The Carbon Efficiency Score serves as a combined metric that evaluates the balance between model performance and environmental impact. A higher efficiency score indicates better utilization of computational resources with lower carbon cost.

Another important feature of the framework is its ability to provide explainable sustainability insights. The system highlights specific training stages where resource usage is inefficient or carbon emissions are disproportionately high. This transparency allows users to understand the direct impact of their training configurations and make informed decisions to improve efficiency.

Furthermore, the integration of a carbon budget control mechanism enhances the practical applicability of the framework. By allowing users to define a maximum allowable carbon emission threshold, the system can dynamically monitor and regulate the training process to ensure that sustainability constraints are not violated. This feature is especially useful in enterprise and cloud-based deployments where cost and environmental regulations are critical considerations.

The framework also incorporates an intelligent recommendation module, which analyzes training patterns and suggests optimization strategies such as reducing epochs, tuning batch sizes, and adjusting learning rates. These recommendations are generated based on real-time data, making them highly relevant and actionable.

Overall, the results clearly demonstrate that CarbonLens AI provides a comprehensive solution for integrating sustainability into machine learning pipelines. The framework not only reduces unnecessary computational overhead and environmental impact but also enhances cost efficiency and decision-making. By embedding Green AI principles directly into the training workflow, CarbonLens AI contributes toward the development of responsible, efficient, and sustainable artificial intelligence systems.

4.2 Result



Figure 4.1: CarbonLens AI Output Dashboard Showing Carbon Profiling, Accuracy–Carbon Tradeoff, and Sustainability Insights

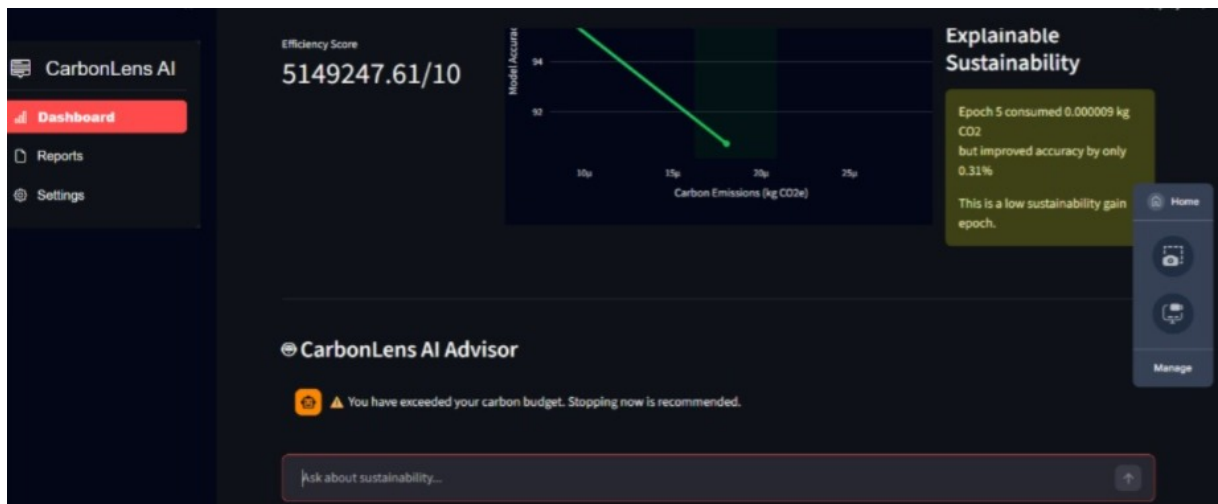


Figure 4.2: AI Chat Advisor Providing Sustainability-Aware Optimization Recommendations

Chapter 5

Conclusion

The proposed mini-project "CarbonLens AI: A Carbon-Aware Sustainability Analytics Framework for Machine Learning Training" successfully addresses the growing need for sustainable and environmentally responsible machine learning practices.

The framework introduces a structured methodology to integrate carbon-awareness into the machine learning training pipeline. By continuously monitoring CPU, GPU, memory utilization, and runtime duration, the system is able to estimate total energy consumption and convert it into carbon emissions using region-based electricity carbon intensity values.

A key contribution of the proposed system is the ability to combine environmental metrics with model performance indicators. Through the generation of the *Accuracy vs. Carbon Tradeoff Curve*, the framework helps identify inefficient training epochs where additional computation results in negligible accuracy gains but significantly increases carbon cost.

The proposed CarbonLens AI framework also provides intelligent optimization suggestions such as:

- early stopping
- batch-size tuning
- reduced epoch recommendations
- carbon budget-aware training

These features promote the concept of Green AI, helping developers balance model performance, cloud operational cost, and environmental sustainability.

Overall, the framework provides a practical, scalable, and academically relevant solution for sustainable machine learning experimentation and responsible AI development.

5.1 Future Work

Although the proposed framework provides strong sustainability analytics capabilities, several advanced improvements can further enhance its real-world applicability.

The future scope of CarbonLens AI includes:

- integration with real-time cloud carbon intensity APIs
- multi-region carbon-aware training deployment
- cloud workload scheduling based on low-carbon zones
- Kubernetes and MLOps integration
- enterprise-level ESG compliance dashboard
- reinforcement learning-based sustainability optimizer
- carbon-aware hyperparameter search
- automated sustainability score benchmarking
- support for distributed GPU clusters
- enterprise audit and reporting automation

These future enhancements can transform CarbonLens AI from an academic mini-project into a deployable industry-grade Green AI sustainability platform.

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